Curriculum Vitae

Jolan Philippe

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1 Curriculum Vitae (short)

Last name:	PHILIPPE
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1.1 Academic background

2019-2022 : **Ph.D. in Computer Science** from IMT Atlantique.
Thesis: "*Contribution to the analysis of the design-space of a distributed transformation engine.*"
Jury:

Chair :	Pr Thomas Ledoux (IMT Atlantique)
Reviewer :	Dr Jesus Sanchez Cuadrado (Universidad de Murcia)
Reviewer :	Pr Matthias Tichy (University of Ulm)
Examiner :	Pr Leen Lambers (Brandenburgische Technische Universität)
Examiner :	Pr Antonio Vallecillo (Universidad de Málaga)
Director :	Dr Gerson Sunye (Université de Nantes)
Supervisor :	Dr Massimo Tisi (IMT Atlantique)
Supervisor :	Dr Hélène Coullon (IMT Atlantique)

2017-2019 : **Master of Science**, *Computer Science*, Northern Arizona University - GPA: 3.67/4
Master thesis : *Systematic Development of Efficient Programs on Parallel Data Structures*, research work done at the *School of Informatics, Computing & Cyber Systems* (Flagstaff AZ, USA) under the supervision of Pr. Frédéric Loulergue.

2016-2017 : **Master degree**, *Computer Science, specialty Master of IT methods applied* *to business management (MIAGE)* from Université d'Orléans (France) — with honors, valedictorian.

2013-2016 : **Bachelor**, *Computer Science* from Université d'Orléans (France) - with honors, valedictorian.

1.2 Professional background

Since September 2025 Associate Professor at Université d'Orléans.

August 2024 - August 2025 Contractual postdoctoral fellow at Université de Rennes - DiverSE team. Two-year contract funded by the project Taranis in the context of PEPR Cloud.

February 2023-July 2024: Contractual postdoctoral fellow at IMT Atlantique - STACK team. One-year contract funded by the ANR project SeMaFoR .
Work carried out in collaboration between IMT Atlantique (STACK team and TASC), Sorbonne University (LIP6 laboratory), and the SMILE group. In addition to the research work around the themes of the SeMaFoR project, I had the opportunity to teach several teaching units, at different branches, as a temporary worker (56h15).

September 2019–December 2022: Contract doctoral student at IMT Atlantique - Naomod and STACK team. Contract funded by the Marie Skłodowska-Curie Actions through the European project Lowcomote, Training the Next Generation of Experts in Scalable Low-Code Engineering Platforms.
Work carried out in collaboration between IMT Atlantique (Naomod team), the University of York, the Universidad Autónoma de Madrid, the University of L'Aquila, the Johannes Kepler Universität Linz, British Telecom, the companies Intex, Uground, CLMS UK and IncQueryLabs. In addition to the research work, I was able to teach a total of 61 hours and 15 minutes to IMT Atlantique students of the FISE engineering training and 10 hours to the Master 2 at the University of Nantes.

September 2017-July 2019: Research assistant at Northern Arizona University - SICCS Software Engineering Research Laboratory.
The job came with the opportunity to take specialized courses to train me in research, refine my skills, and deepen my understanding of the distributed systems field.

November 2016-August 2017: Part-time research engineer at the Université d'Orléans - LMV team. Contract funded by the Girafon project.

April 2016-August 2016: Intern, at Acatus / Atimic.

The internship consisted of testing solutions implemented in J2EE, in particular by handling the Spring framework.

1.3 Teaching [1]

2026 **Associate professor at Université d'Orléans (France)**
- PW in *Programmation N-tiers*, Bachelor 3 computer science (30h)

2025 **Associate professor at Université d'Orléans (France)**
- L in the *Initiation to research*, Master 1 ARIAS (4h30)
- PW in the *Web services* module, Master 1 MIAGE (36h)
- L/PW in the *Introduction to DevOps* module, Master 1 MIAGE ALT (14h)
- Support for projects in the *Projet* module, Master 1 MIAGE FI (40h)

Temporary worker at ISTIC - Rennes (France)
- PW in the *Software Engineering* class, Bachelor 2 computer science (30h)

2024 **Temporary worker at ESIR - Rennes (France)**
- PW in the *Object-Oriented Programming* class, 1^{st} year of engineering diploma cycle (28h)

Temporary worker at IMT Atlantique - Nantes (France)
- L/PW in the *Object-Oriented Programming* class, Master 1 MOST (10h)

2023 **Temporary worker at IMT Atlantique - Nantes (France)**
- L/PW in the *Services* module 3^{rd} year of engineering diploma cycle (22h30)
- L/PW in the *Distributed architectures* module, 1^{st} year of engineering diploma cycle, training by apprenticeship (18h45)
- PW in the *Algorithmics and discrete mathematics* module, 1^{st} year of engineering diploma cycle (replacement, 5h)

2022 **Temporary worker at IMT Atlantique - Nantes (France)**
- PW in the *Algorithmics and discrete mathematics* module, 1^{st} year of engineering diploma cycle (20h)

2021 **Temporary worker at IMT Atlantique - Nantes (France)**
- PW in the *Databases and Interactive Software* module, 1^{st} year of engineering diploma cycle (18h45)

2020 **Temporary worker at IMT Atlantique - Nantes (France)**
- PW in the *Algorithmics and discrete mathematics* module, 1^{st} year of engineering diploma cycle (22h30)

Temporary worker at Université de Nantes (France)
- Support for research projects in the *Capstone* module, Master 2 in computer science (10h)

[1] A more detailed description will follow this CV. L: lecture, PW: practical work (lab).

1.4 Supervision

From Nov. 2026 **Nathan Fouéré, Ph.D candidate, Taranis project**
- Empirical and formal study of the resilience of decentralized Infrastructure-as-Code solutions.

May 2026 - Aug 2026 **Paul Nicolas, internship Master 1 computer science, UniVerSE project**
- Exploration of UVL (Universal Variability Language) ecosystem with its different semantics.

From Sept. 2025 **Olivia Proust, Ph.D candidate, For-Coala project**
- Formalization and verification of Infrastructure as Code (IaC) scripts for configuration management (Ansible).

From Sept. 2024 **Haitam El Hayani, Ph.D candidate, Taranis project**
- Extending Infrastructure as Code (IaC) languages, focusing on enhancing their capabilities and improving their integration with modern cloud deployment practices.

1.5 International publications[2]

- 1 publication in international journal (JSS)
- 7 publications in international conferences (SAC'19, HPCS'19, PDCAT'19, ICA3PP'19, SLE'21, SANER'24, ADBIS'26)
- 6 publications in international workshops (MODELS'20, FTfJP'23, ICE'24, MODEVAR'26, ConfLang'26, ConfLang'26)

1.6 Languages

I have a good professional and scientific level in English. I spent more than two years at Northern Arizona University, in Flagstaff, USA, as part of my research engineering contract and obtaining my Master of Science. During this period, the main language of communication was English. Likewise, meetings and communications with members of the STACK and DiverSE teams are mainly in English.
French: native; Spanish: school level.

[2] A more detailed description will follow this CV.

2 Teaching activities

2.1 Teaching synthesis

Since September 2020, I have given a total of 321 hours 21 minutes of teaching, equivalent to 303.05 practical work hours (EQTD). Details of these hours are given below.

Table 3 presents the list of all the teachings in which I have intervened. The abbreviations used regarding the courses and schools are detailed in Table 1 and Table 2. About levels, we use B1-B2-B3 for bachelor years, M1-M2 for master years, and Y1-Y2-Y3 for engineering cycle years.

MIAGE	Computer methods applied to business management • FI: Initial Training • ALT: Apprenticeship program
ARIAS	Distributed applications, artificial intelligence and security
MOST	Management and Optimization of Supply Chains and Transport
FISE	Engineering diploma cycle
FIL	Engineering diploma cycle, training by apprenticeship
LOGIN	Software engineering and innovation
ALMA	Computer Science, major in software architecture
CS	Computer Science

Table 1: Used abbreviations for school majors

UO	University of Orléans
ISTIC	Institute of Information and Communication Technologies (Rennes)
ESIR	Engineering school of Rennes
IMTA	Mines-Télécom Institute - Atlantique (Nantes)
NU	Nantes University
NAU	Northern Arizona University

Table 2: Used abbreviations for school names

2.2 Details about teaching at University of Orléans

N-Tiers Programming - B3 CS - [Support]

This course aims to introduce the MVC (Model-View-Controller) pattern into application design. Taught in Java, the course uses two technologies to separate the view from the controller. First, JSP/Servlets are used as a low-level technology to understand the concepts of dynamic views, templates, and request handling. Then, using the Spring Boot framework, views are implemented as Thymeleaf templates, while the controller exposes endpoints through a REST API. This course serves as an introduction to Web Service APIs, which are covered in the first year of the MIAGE Master's program (M1 MIAGE).

Label	Year	School	Hours	Level	Stdts
As associate professor					
N-Tiers Programming	2026	UO	PW, Exam. (30h, 30 EQTD)	B3 CS	18
Initiation to research	2025	UO	Lec., Exam. (4h30, 6.75 EQTD)	M1 ARIAS	31
Web services	2025	UO	PW, Exam. (36h, 36 EQTD)	M1 MIAGE	40
Introduction to DevOps	2025	UO	Lec., PW, Exam. (14h, 16 EQTD)	M1 MIAGE ALT	30
Project	2025	UO	Project (39.6 EQTD)	M1 MIAGE FI	33
As temporary teacher					
Software Engineering	2025	ISTIC	PW (30h, 20 EQTD)	ISTIC L2	20
Object Oriented Programming	2024	ESIR	PW (26h, 17.33 EQTD)	ESIR Y1	17
Object Oriented Programming	2024	IMT Atlantique	Lec., PW (10h, 13 EQTD)	M1 MOST	10
Services	2023	IMT Atlantique	Lec., PW, Exam. (22h30, 21.25 EQTD)	FISE Y3 LOGIN	22
Distributed Architectures	2023	IMT Atlantique	Lec., PW, Exam. (28h45, 20 EQTD)	FIL Y1	30
Algorithmic and discrete mathematics	2020	IMT Atlantique	PW, Exam. (22h30, 22.5 EQTD)	FISE Y1	22
	2022	IMT Atlantique	PW, Exam. (20h, 14 EQTD)	FISE Y1	22
	2023	IMT Atlantique	PW, Exam. (5h, 5 EQTD)	FISE Y1	22
Databases and Interactive Software	2021	IMT Atlantique	PW, Exam. (18h45, 27.87 EQTD)	FISE Y1	22
Research project (Capstone)	2020	Univ. Nantes	Project (10h, 10 EQTD)	M2 ALMA	4
Principle of languages	2019	NAU	Cours, TP (3.75 EQTD)	B3 CS	30

Table 3: Teaching provided as associate professor and as temporary teacher

Initiation to research - M1 ARIAS - [Support]

The "Initiation to Research" course offered in the first year of the ARIAS Master's program aims to introduce the different research teams of LIFO (Laboratoire d'Informatique Fondamentale d'Orléans), namely CA (Constraints and Learning), GAMoC (Graphs, Algorithms and Models of Computation), Pamda (Parallelism and Data Management), LMV (Languages, Models and Verification), and SDS (Data and Systems Security). Each team freely presents research topics, methods, and exercises related to its areas of expertise. For my part, I provided a general introduction to the LMV team, followed by presentations on program verification, in particular the Curry-Howard correspondence, in the context of system reconfiguration.

Web Services - M1 MIAGE - [Support]

The "Web Services" course teaches students about REST APIs and introduces distributed services. It begins with writing HTTP requests to an existing API before moving on to developing REST endpoints and Java clients using Spring Boot. Students then design complete REST APIs and make them communicate with each other, while progressively introducing security mechanisms using JWS. Finally, the course broadens their knowledge of distributed service technologies by presenting gRPC, as well as the concepts of IDLs, proxies, and stubs, still within the Spring Boot environment.

Introduction to DevOps - M1 MIAGE ALT - [Support]

The "Introduction to DevOps" course is taught to first-year MIAGE Master's students enrolled in a work-study program. It introduces the main principles of DevOps, including its philosophy, objectives, and challenges, as well as the concepts of Continuous Integration and Continuous Delivery (CI/CD). For CI, the course covers Test-Driven Development

(TDD) and automated testing. For CD, it introduces Infrastructure as Code (IaC) and several widely used technologies, including Docker, Ansible, Terraform, and Kubernetes.

Project - M1 MIAGE FI - [Support]

The project module is a direct practical application of the Web Services course. Working in teams of four or five, students are asked to design and develop a multi-service application using REST APIs. Three deliverables are expected, one per month, to ensure regular monitoring of the project and steady progress throughout the module. Project topics are varied and may involve the development of an application, a software package, or even a game. As this module is only offered to students in the initial education program, additional learning sessions on application deployment, including Docker and Terraform, are provided. Assessment takes the form of oral presentations, during which I invite an industry professional to provide an external perspective on the students' working methods.

2.3 Details about teaching at Institut supérieur des technologies en informatique et communication (ISTIC)

Software Engineering - ISTIC B2 - [Support]

The "Software Engineering" course for second-year bachelor students at ISTIC aims to introduce the principles and practices required for collaborative software development. Its main objective is to expose students to real-world software development challenges and equip them with the tools and methods to address them effectively.

The main topics covered in the course include: (i) Fundamentals of software architecture, with an emphasis on the modular organization of applications using classes, interfaces, and modules; (ii) Challenges of collaborative work, including the management of development teams of approximately 10 people, communication, work organization, and code sharing; (iii) Software maintenance and evolution, including the modification of existing code, documentation, and code readability; (iv) Software testing, including unit and integration testing; (v) Modern development tools, such as integrated development environments (VS Code), version control systems (e.g., Git), and the use of artificial intelligence tools (LLMs such as ChatGPT or Copilot) for programming; (vi) Agile methodologies, enabling iterative and flexible development adapted to changing requirements.

The course combines theoretical lectures, tutorials, and practical sessions. Students take part in collaborative projects designed to simulate real-world software development conditions. They apply the concepts introduced during the course, such as version control, documentation, and test writing, while also developing their algorithmic problem-solving and programming skills.

I supervise the practical sessions and projects, in which students explore these concepts through hands-on exercises. I am also responsible for invigilating and grading assessments.

2.4 Details about teaching at Ecole Supérieure d'Ingénieurs de Rennes (ESIR)

Object Oriented Programming - ESIR Y1 - [Support]

The "Object-Oriented Programming in Java" course at ESIR is aimed at first-year engineering students, with the primary focus on introducing and deepening object-oriented programming (OOP) concepts. The key topics include: (i) Fundamental principles of OOP: encapsulation, inheritance, polymorphism, and abstraction; (ii) Mechanisms and functionality of OOP concepts: classes, late binding, and polymorphism; (iii) Standard data structures; (iv) Practical implementation of these concepts in Java. Throughout the course, a combination of theoretical lectures, tutorials, and practical lab sessions are planned to enhance the students' understanding. I am responsible for supervising the hands-on labs, where students will apply the theoretical knowledge in coding exercises and real-world scenarios. In addition, I will oversee and grade the exams, ensuring students grasp the critical aspects of OOP and its practical application in Java.

2.5 Details about teaching at IMT Atlantique

Object Oriented Programming - M1 MOST - [Support]

This course serves as an introduction to object-oriented programming for MOST master's students. In the 1^{st} year, this master's degree aims to provide training in the fundamentals of supply chain management and optimization. The courses in this master's degree are entirely given in English. This course is part of a larger module around operational research. This module includes two other courses, around which object-oriented programming is structured: meta-heuristics and a TSP lab. The object-oriented programming course introduces the notions of classes, instances, inheritance, encapsulation, and polymorphism. I created the course material, and the practicals are carried out in Python, which explains the absence of other notions such as interfaces or anonymous classes.

Distributed architectures - FIL Y1 - [Support] / Services - FISE Y3 LOGIN - [Support]

The "Distributed Architectures" course and the "Services" course share the same educational content. These courses were structured in a similar way over the first three sessions, with a combination of lectures, tutorials and practices to deepen the concepts covered. The key concepts covered were REST APIs, GraphQL and gRPC. I was responsible for presenting the lecture on gRPC for the two modules: "Distributed Architectures" and "Service". The following sessions were devoted to finalizing practical work through a project for managing cinema screening reservations for a user. Finally, the last session was dedicated to the presentation of the projects carried out by the students. In collaboration with the module manager, I was responsible for evaluating the students' oral presentations as well as the code produced during these projects.

Algorithmic and discrete mathematics - FISE Y1 - [Support]

The course was structured around coherent practical sessions. Before each session, students had to familiarize themselves independently with the concepts to be covered. The sessions began with a brief assessment of the course content to check their understanding. Then the sessions involved the use of the game PyRat to put into practice the algorithms studied, in particular to move a character in a maze, offering a concrete vision of the application of graph algorithms in Python. These algorithms included graph traversal methods, Dijkstra's algorithm for shortest paths, solving the traveling salesman problem to optimize routes, and the introduction of greedy and heuristic concepts. The final session allowed students to create their own strategy based on previous learning, thus serving as a final assessment for the course. This combined approach of self-study and practice in PyRat provided real-world experience promoting a deep understanding of algorithmic concepts and their application in real-world contexts. I supervised these practical work sessions and evaluated the students' oral presentations as well as the code produced during these projects.

Databases and Interactive Software - FISE Y1 - [Support]

The "Database and Interactive Software" course was divided into two parts: the first focused on databases, while the second covered human-machine interfaces (HMI). I took charge of the practical work relating to databases. The main objective was to raise students' awareness of information systems and their structures. During this course, students learned about SQL query language, normalization theory, and conceptual modeling. In addition to the classroom sessions, the students completed a project where they had to model a database for an email application, test it and perform queries. My participation was focused on supervising these practical work sessions relating to databases.

2.6 Details about teaching at Université de Nantes

Projet de recherche (Capstone) - M2 ALMA

I supervised a group of four students as part of the Capstone research project module by organizing bi-monthly meetings. The research project focused on the implementation of OCL (Object Constraint Language) using Scala and Spark. During these regular meetings, we discussed the progress of the project, identified the challenges encountered and developed strategies to overcome them. My role was to provide technical support and guide the members of the group in their research.

Private lessons and academic support

I have had the privilege of supporting pupils and students at different academic levels, particularly in mathematics for middle and high school levels (Baccalaureate in Science). In addition, I provided specific support in statistics to 3rd-year Bachelor's degree and

1st-year Master's degree psychology students at the University of Nantes. Furthermore, I also contributed to the understanding of the functional specifications for 1st year's Master degree MIAGE students.

3 Research activities

3.1 Research conducted during my Master's degree (November 2016 - July 2019)

My research during my Master's degree first took place during the first year (M1) of my MIAGE Master's program (Methods in Computer Science Applied to Business Management) at the University of Orléans, within the Girafon project. This project was funded by the Centre-Val de Loire Region through an academic research project call (APR-IA). The second part of this work was funded through a 'graduate assistant" scholarship during my two years at Northern Arizona University. All this research was supervised by Frédéric Loulergue, while he was a professor at Northern Arizona University. As part of my 'Master of Science" degree, all the work presented here was developed in a Master's thesis [3], whose scope was comparable to that of a short doctoral dissertation.

Supervisor :	Pr Frédéric LOULERGUE (Northern Arizona University)
Examiner :	Dr Hélène COULLON (IMT Atlantique)
Examiner :	Dr Frédéric DABROWSKI (Université d'Orléans)
Examiner :	Dr Michael GOWANLOCK (Northern Arizona University)

General topic and context. The increase in data volumes ("Big Data") requires increasingly efficient computing techniques. Modern parallel architectures (e.g., clusters of machines) are systems containing several processors working simultaneously. The modularity of computing resources makes it possible to reduce overall computation time by distributing tasks among processors, while also enabling scalability by increasing the number of processors. Big Data applications often process sensitive data (e.g., in healthcare or epidemiology), which creates a need for the development of correct Big Data applications.

Keywords. Functional programming, Parallelism, Bulk Synchronous Parallelism, Formal verification, Algorithmic skeletons, Python, Coq

Scientific challenges and adopted approach. Writing correct parallel programs is challenging because of the complexity introduced by concurrency management and task distribution. Algorithmic skeletons are computational patterns equipped with parallel implementations and provide a structured solution to this problem. However, applying them to complex data structures, such as trees and graphs, raises major challenges in terms of formalization, correctness, and efficiency. To address these issues, we adopted a mixed

[3] https://jolanphilippe.github.io/docs/publications/2019_nau_philippe.pdf

approach combining the rigorous mathematical formalization of skeletons using the Coq proof assistant, thereby guaranteeing their correctness, with the practical implementation of these concepts as software libraries targeting modern computing environments (i.e., Python).

Summary of contributions. A first contribution focused on the formalization of skeletons applied to non-linear structures such as binary trees, including a formal proof of execution equivalence for patterns such as *map*. This work, carried out using the Coq proof assistant, extended SyDPaCC, an existing Coq library for the systematic development of programs for parallel and Cloud computing. In addition to integrating a formalization of binary trees and skeletons, we extended the notion of equivalence between compositions of sequential functions and compositions of parallel functions, including function equivalences involving intermediate types (e.g., serializing a tree as a list). A second contribution was the creation of PySke, a Python library dedicated to the parallel manipulation of lists and trees (binary or otherwise). PySke provides two execution modes: a "standard" execution mode, using Python as a dynamic scripting language, and a lazy execution mode intended to optimize successive skeleton calls. From the user's perspective, PySke follows a SIMD (Single Instruction Multiple Data) approach, making it easier to use, whereas its implementation relies on MIMD (Multiple Instruction Multiple Data) for performance. PySke includes numerous skeletons, both for manipulating data distribution and for implementing patterns such as *map*, *reduce*, *fold-left*, and *fold-right*. The work described here resulted in four publications in peer-reviewed international conferences, three *extended abstracts* accompanied by posters, and a Master's thesis defended before an international committee.

3.2 PhD research (September 2019 - December 2022)

This PhD research was funded by the European Lowcomote project through the European Union's Horizon 2020 research and innovation programme under a Marie Skłodowska-Curie grant agreement.

Chair :	Pr Thomas Ledoux (IMT Atlantique)
Reviewer :	Dr Jesus Sanchez Cuadrado (Universidad de Murcia)
Reviewer :	Pr Matthias Tichy (University of Ulm)
Examiner :	Pr Leen Lambers (Brandenburgische Technische Universität)
Examiner :	Pr Antonio Vallecillo (Universidad de Málaga)
Director :	Dr Gerson Sunye (Université de Nantes)
Supervisor :	Dr Massimo Tisi (IMT Atlantique)
Supervisor :	Dr Hélène Coullon (IMT Atlantique)

General topic and context. Model-Driven Engineering (MDE) is a software engineering approach that places models at the center of the design process. It encompasses a variety of modeling techniques and operations. Among these, model transformation is a complex operation with multiple implementations based on different semantics, paradigms, technologies, and design choices. These fundamental differences make it difficult to compare all existing approaches. However, it is clear that each solution was designed for a particular purpose, with constraints related to its use case or execution environment.

Keywords. Model-Driven Engineering, Software Engineering, Apache Spark, Model transformations, Model queries, Correctness, Characterization

Scientific challenges and adopted approach. The main difficulties involved in model transformation concern managing the complexity and diversity of "Very Large Models" (VLMs) in a distributed computing context. These models, which may represent extremely large datasets, require tools that satisfy scalability, efficiency, and transparency requirements. The heterogeneous nature of existing approaches, whether based on data parallelism, task parallelism, or asynchronous execution, complicates their comparison and limits the development of unified solutions. Furthermore, the impact of engineering choices, both in transformation engine design and in their configuration, remains insufficiently explored, even though these decisions directly affect performance. To address these challenges, I adopted a systematic methodology combining parallelization strategies tailored to the use case (e.g., MapReduce, Pregel, distributed OCL), rigorous evaluation of existing engines (e.g., ATL, ETL), comparison of multiple execution semantics using formal methods (equivalence proofs), and exploration of the impact of configurations on performance (i.e., computation time), with the aim of designing a distributed solution providing a high degree of configurability for model transformations.

Summary of contributions. A first contribution of this work consisted in the exploration and **implementation in Scala of different distributed programming models**, such as Spark (distributed pure functional programming), MapReduce, and Pregel, to execute queries on models using expressions from the OCL language. These paradigms improve the performance and scalability of queries used in endogenous or exogenous transformations on large models containing several thousand elements. By relying on these distributed models, this approach reduces the need to develop rigid unified solutions by providing flexibility in the choice of transformation paradigm, thereby enabling various scenarios to be handled efficiently while reducing the complexity of the proposed solutions.

The second contribution focuses on the design of distributed transformation engines. It relies on a **refined specification of CoqTL** (a transformation language written using the Coq proof assistant), first through a new specification increasing the use of intermediate results, and subsequently through a specification using two successive map-reduce phases. Both specifications were **formally validated through equivalence proofs**, and the latter was then implemented on Spark in a prototype called SparkTE. This approach guarantees consistency of the produced results while providing better parallelization and reduced computation times. The formalization of transformation processes and the validation of the proposed solutions therefore provide solid foundations for the development of distributed transformation engines without relying on generic solutions that are often poorly suited to specific cases, while enabling precise modeling tailored to each particular need.

However, the Spark implementation relies on a manual extraction of Coq code to Scala, followed by the replacement of data structures with distributed data structures. In order to **guarantee the semantic preservation of the engine in the Spark code, we formalized part of Spark in Coq**, enabling the use of Scallina (a Coq-to-Scala extractor) to generate Spark code directly from the Coq environment. This formalization strengthens consistency between the different layers of the solution by aligning development tools with formal requirements, thereby reducing the risk of semantic divergence when managing large-scale transformations.

Finally, a third contribution introduces **adaptive configurations** for SparkTE, making the engine configurable. These configurations enable fine-grained performance optimization according to the characteristics of transformations and execution environments. The ability to adapt the engine to different execution contexts addresses the need to limit the proliferation of heterogeneous technical solutions and configurations while maximizing efficiency through a flexible approach tailored to each use case.

These contributions are illustrated through experiments on several use cases well established within the community (e.g., the *Transformation Tool Contest* of the STAF conference), demonstrating the flexibility, efficiency, and scalability of the proposed solutions in addressing the challenges raised by VLMs. They therefore provide an answer to the limitations of overly rigid unified solutions by introducing configuration granularity and adaptability that are essential for managing the complexity of transformations over very large models.

The work described in this PhD resulted in one publication in a peer-reviewed international conference, two publications in peer-reviewed international workshops, three research reports (project deliverables) targeting an international audience, and one summary of results presented at a local event.

3.3 Postdoctoral research (February 2023 - August 2025)

The research described below was conducted around a common topic, the administration of distributed systems, during two postdoctoral contracts. I deliberately group the work conducted during these two contracts into a single section and under the same heading. Although this research was conducted in different contexts and with different teams, both projects followed the same research direction, and the work carried out during the second contract is a logical continuation of the first. The first contract was conducted in Nantes as part of the ANR SeMaFoR project (Self-Management of Fog Resources), and the second in Rennes as part of the Taranis project within the PEPR Cloud programme. Beyond the "postdoctoral" job title, I consider this period as a bridge between my PhD research and my goal of becoming an Associate Professor.

General topic and context. The reconfiguration of distributed systems is a practice that ensures the flexibility, availability, and resilience of modern infrastructures (Cloud, Fog, or Edge). The DevOps paradigm has emerged to ensure reliable maintenance (e.g., updates) of these systems without service interruption. This has led to the role of the DevOps engineer. This approach aims to unify and automate the methods used to ensure continuous integration and continuous deployment of applications. Among the mechanisms supporting automation are declarative approaches, in which the desired state of resources (infrastructure or applications) is specified rather than the procedure for reaching that state. The declarative engine is then responsible for synthesizing a reconfiguration plan containing the required modifications and subsequently executing it. This mechanism is, for instance, widely adopted in *Infrastructure-as-Code* (IaC), a field in which infrastructure configurations and application deployments are specified as computer programs that can be shared, versioned, and tested in the same way as traditional software.

Keywords. DevOps, Infrastructure-as-Code, Reconfiguration, Model checking, Constraint programming

Scientific challenges and adopted approach. The reconfiguration of distributed systems raises major scientific challenges, particularly when the architecture requires decentralized approaches. In environments such as Fog Computing or Service-Oriented Architectures, each node is designed to operate autonomously and to share as little information as possible with other nodes, despite functional or structural dependencies. This constraint requires complete decentralization of the reconfiguration processes, where each node executes a local reconfiguration program that may be synchronized with other nodes through synchronization barriers. In a decentralized setting, the objective is to define, for each node, a reconfiguration either as a target state to be reached or as a sequence of operations to be executed. However, several challenges arise in this context: (i) incompatibilities between reconfiguration programs (e.g., two incompatible versions of a resource or service may be imposed on dependent nodes), (ii) local planning while respecting constraints related to the global topology, (iii) maintaining performance (scalability, speed) despite decentralization, and (iv) limiting information sharing. In a more general context, whether centralized or decentralized, IaC languages such as HCL, used by Terraform, are predominantly declarative. This approach facilitates the definition of target states for resources but introduces additional challenges, such as guaranteeing certain properties (e.g., resource availability or security levels). Furthermore, because of their simplicity, the tooling surrounding these languages mainly provides syntactic support. Beyond checking that field values use the appropriate types, it remains difficult to guarantee to users that their configurations are suitable for the type of application being deployed (e.g., resource sizing issues).

Contributions. This research led to the development of **Ballet, a decentralized reconfiguration tool for a cross-DevOps context**, that is, an environment in which several DevOps engineers work on disjoint parts of an infrastructure. Ballet consists of two main modules: a reconfiguration *planner*, which generates plans from component life cycles, and an execution engine that applies these plans to the nodes of the system.

At the core of Ballet is the Concerto reconfiguration model, which makes it possible to specify reconfiguration scripts according to precise behaviors (e.g., update, deployment, shutdown) as controllers resembling interconnected Petri nets. This approach offers two advantages: (i) within a node, it makes it possible to define a partial order among operations, thereby enabling fine-grained concurrency between them; and (ii) across nodes, it captures dependencies and imposes the synchronization required to guarantee system consistency.

Ballet's planner relies on two key concepts: (i) **local constraint solving**, using a SAT solver to guarantee correct solutions for the specified reconfiguration objectives; and (ii) **node-to-node dissemination of synchronization constraints**, based on the generated programs. Ballet's **decentralized execution engine**, named Concerto-D, uses the generated reconfiguration plans to carry out the system reconfiguration. In addition, Concerto-D was formally defined and implemented in the Maude *model checker*. This formalization provides a mechanized solution for formally verifying Concerto-D and for providing guarantees regarding the validity of reconfiguration processes.

However, Ballet did not address cases where a reconfiguration became unsatisfiable because of conflicts between objectives expressed by different DevOps teams. To address this limitation, we developed Ballet$^+$, **an extension introducing explicit conflict management**. Ballet$^+$ relies on the QuickXplain algorithm to identify and explain minimal sets of conflicting constraints. This approach allows teams to understand the causes of a conflict and to adapt their reconfiguration requests accordingly. **Conflict explanations are efficiently propagated through Ballet's constraint dissemination mechanism.**

The performance of Ballet and Ballet$^+$ was evaluated on complex infrastructures such as OpenStack, demonstrating significant improvements. Ballet achieved a 40

In future work, **Ballet will be integrated into SeMaFoR, a collaborative and consensus-based decision-making process for Fog environments**. SeMaFoR will integrate an automatic decentralized coordination mechanism for reconfiguration based on MAPE-K autonomic loops.

In parallel, I also contributed to the formalization of reconfiguration planning in a centralized context through **a formal model of Terraform**. This contribution improved our understanding of ambiguities in execution plans generated by Terraform and enabled the exploration of **safety guarantees for IaC deployments using provisioning solutions**. More specifically, I participated in the discussions and experiments that led to the formalization of an operational model of Terraform in the Maude *model checker*. This work introduced µProvisioner, a mechanized formalization for analyzing and verifying the safety of Terraform deployments using model-checking techniques. Our contributions include the analysis of ambiguous execution orders, the study of implicit dependencies between resources, and the evaluation of the model's validity on both real-world and synthetic cases. This work resulted in a tool, *BPlan*, which provides an enriched infrastructure reconfiguration plan for Terraform.

These contributions are part of a broader approach aimed at improving infrastructure management through formal planning techniques, both in centralized environments with

Terraform and in decentralized environments with Ballet and Ballet$^+$.

3.4 Research since September 2025

Since September 2025, my research has continued my previous work on the reconfiguration of distributed systems, with an increased emphasis on verifying configurations and the mechanisms that make them evolve. The common objective is to make the guarantees associated with a configuration or its replacement explicit and verifiable, whether these concern Infrastructure-as-Code, configuration spaces, or data-processing systems.

3.4.1 Taranis

Since joining the University of Orléans, I have strengthened my involvement in the Taranis project within the PEPR Cloud programme, in collaboration with Olivier Barais, Benoît Combemale, and Stéphanie Challita from IRISA in Rennes, particularly through the PhD research of Haitam El Hayani.

General topic and context. Infrastructure-as-Code (IaC) languages have become central to automating the deployment and administration of Cloud infrastructures. However, their increasing complexity introduces quality issues that may affect security, maintainability, performance, or the behavior of deployed applications. My research within Taranis focuses on improving quality-assurance mechanisms for these configurations.

Keywords. Infrastructure-as-Code, Cloud, Quality assurance, DevOps, Static and dynamic analysis

Scientific challenges and adopted approach. The main issue is to move beyond syntactic and static verification of configurations in order to account for their behavior in production. We adopt an approach combining a systematic characterization of existing defects and analysis techniques with the use of runtime information to guide the analysis and evolution of configurations.

Contributions. A first contribution is a systematic literature review on quality assurance for IaC languages. It proposes a taxonomy of existing issues and techniques and notably highlights the limited number of studies addressing infrastructure performance and dynamic behavior. Building on this observation, we proposed POLIAC, an approach combining an IaC configuration, requirements expressed in natural language, and operational data. POLIAC uses LLMs to suggest infrastructure modifications consistent with the observed behavior of the deployed system.

3.4.2 UniVerSE

I lead the UniVerSE project (*Unification and Verification of configuration SpacEs*), which was submitted to the ANR under the JCJC call, with support from the ICVL.

General topic and context. Configuration spaces represent the different valid configurations of a software system. The Universal Variability Language (UVL) aims to provide a common language for representing these spaces, but its various implementations currently rely on semantic choices that are not always consistent or made explicit.

Keywords. Configuration spaces, Software variability, UVL, Formal semantics, Verification

Scientific challenges and adopted approach. The objective of UniVerSE is to provide common formal foundations for the specification and verification of configuration spaces. A first step consists in identifying and making explicit the semantic variations currently found within the UVL ecosystem, and then formalizing well-formedness and typing properties and, ultimately, the semantics of the language. We favor a mechanized approach that also allows these definitions to be executed on concrete UVL models.

Contributions. We developed *mech-uvl*, a mechanized environment for UVL. Its formal core, developed in Dafny, defines, among other elements, the abstract syntax, well-formedness and typing conditions, as well as associated executable checkers. It also makes explicit several semantic variation points among existing implementations. A C# front-end makes it possible to directly apply these analyses to UVL files, thereby providing an initial foundation for the development of a mechanized formal semantics of the language.

3.4.3 RaDyD

I co-lead the RaDyD project (*Resilience through Dynamic Adaptation for Data Processing*) in collaboration with the PAMDA team at LIFO. The project received BQR (*Bonus Qualité Recherche*) funding from the University of Orléans and is supported by the CIOM.

General topic and context. Modern data-processing systems increasingly rely on interchangeable software and hardware components. This composability makes it possible to adapt a system when certain resources become unavailable or constrained, but such reconfiguration may also alter the semantics or quality of the produced result.

Keywords. Resilience, Dynamic reconfiguration, Data systems, Semantic contracts, Formal methods

Scientific challenges and adopted approach. RaDyD aims to exploit reconfiguration as a resilience mechanism while controlling its semantic consequences. We propose associating substitutions between configurations with contracts that distinguish exact, conditionally equivalent, or approximate transformations with explicit bounds. The objective is therefore to move from an ad hoc fallback mechanism toward controlled and explainable degradation that can ultimately be mechanically verified.

Contributions. A first contribution, published at ADBIS, introduces the concept of semantics-aware resilience for composable data systems. This work frames the problem in terms of semantic substitutability, proposes a contract-guided reconfiguration architecture, and identifies the main scientific challenges related to contract specification, decision-making, evaluation, and mechanized verification.

3.4.4 ForCoala

I participate in the ANR For-Coala project, in particular in collaboration with Hélène Coullon from IMT Atlantique and through the PhD research of Olivia Proust.

General topic and context. For-Coala aims to provide formal guarantees for Infrastructure-as-Code languages used to configure and reconfigure distributed systems. In particular, the project seeks to bridge widely used practical tools, such as Ansible, and academic languages equipped with formal foundations, with the long-term objective of enabling the verification of programs and transformations between languages.

Keywords. Infrastructure-as-Code, Ansible, Formal semantics, Static analysis, Verification

Scientific challenges and adopted approach. A major difficulty arises from the sometimes implicit or counter-intuitive semantics of industrial IaC languages. Ansible, in particular, has complex rules governing variable scope and precedence. Our approach first consists in precisely characterizing these behaviors and developing analyses capable of making them explicit, thereby progressively providing the foundations required for their formalization and verification.

Contributions. A first contribution is *Scopeannon*, a static analyzer dedicated to variable-scope ambiguities in Ansible. This work characterizes variable resolution by accounting, in particular, for precedence, dynamic availability, role ordering, and lazy evaluation. Scopeannon then makes this resolution explicit in order to identify potentially unexpected scope selections or variable overrides.

3.5 List of publications from 2017 to 2026

These publications are part of the themes that I was able to address during my research work. The order of authors represents the share of contribution of each, except for project deliverables. For the latter, the authors all contributed equally; the order was decided arbitrarily.

3.5.1 International revues with published proceedings

1. Haitam El Hayani, Jolan Philippe, Stéphanie Challita, Olivier Barais, and Benoît Combemale. “Quality assurance in infrastructure as code: Issues, approaches, and open challenges”. In: *Journal of Systems and Software* (2026), p. 113080. ISSN: 0164-1212. DOI: https://doi.org/10.1016/j.jss.2026.113080. URL: https://www.sciencedirect.com/science/article/pii/S0164121226003134
   Note : Q1-rank revue

3.5.2 International conferences with published proceedings

1. Frédérique Loulergue and Jolan Philippe. “Towards Semantics-Aware Resilience in Composable Data Systems”. In: *The 30th European Conference on Advances in Databases and Information Systems (ADBIS 2026)*. 2026
   Note : B-rank conference

2. Jolan Philippe, Antoine Omond, Hélène Coullon, Charles Prud’Homme, and Issam Raïs. “Fast Choreography of Cross-DevOps Reconfiguration with Ballet: A Multi-Site OpenStack Case Study”. In: *2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. Rovaniemi, Finland: IEEE Computer Society, Mar. 2024. DOI: 10.1109/SANER60148.2024.00007
   Note : A-rank conference, Acceptance : 25.6% (244 submissions)

3. Jolan Philippe, Massimo Tisi, Hélène Coullon, and Gerson Sunyé. “Executing Certified Model Transformations on Apache Spark”. In: *Proceedings of the 14th ACM SIGPLAN International Conference on Software Language Engineering*. SLE 2021. Chicago, IL, USA: Association for Computing Machinery, 2021, pp. 36–48. URL: https://doi.org/10.1145/3486608.3486901
   Note : B-rank conference, Acceptance : 47% (44 submissions)

4. Frédéric Loulergue and Jolan Philippe. “Automatic Optimization of Python Skeletal Parallel Programs”. In: *Algorithms and Architectures for Parallel Processing*. Ed. by Sheng Wen, Albert Zomaya, and Laurence T. Yang. Cham: Springer International Publishing, 2020, pp. 183–197. URL: https://doi.org/10.1007/978-3-030-38991-8_13
   Note : B-rank conference, Acceptance : 29% (251 submissions)

5. Frédéric Loulergue and Jolan Philippe. “New List Skeletons for the Python Skeleton Library”. In: *2019 20th International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT)*. 2019, pp. 392–397. URL: https:

//doi.org/10.1109/PDCAT46702.2019.00077
Note : B-rank conference, Acceptance : 30.7% (257 submissions)

6. Jolan Philippe and Frédéric Loulergue. “PySke: Algorithmic Skeletons for Python”. In: *2019 International Conference on High Performance Computing and Simulation (HPCS)*. 2019, pp. 40–47. URL: https://doi.org/10.1109/HPCS48598.2019.9188151
Note : B-rank conference, unavailable acceptance rate.

7. Jolan Philippe and Frédéric Loulergue. “Parallel Programming with Coq: Map and Reduce Skeletons on Trees”. In: *Proceedings of the 34th ACM/SIGAPP Symposium on Applied Computing*. SAC ’19. Limassol, Cyprus: Association for Computing Machinery, 2019, pp. 1578–1581. URL: https://doi.org/10.1145/3297280.3299742
Note : A-rank conference, Acceptation : 24.2% (1067 submissions)

3.5.3 International workshops with published proceedings

1. Olivia Proust, Hélène Coullon, Jolan Philippe, and Frédéric Loulergue. “Scopeannon: A Static Analysis of Ansible’s Scopes Ambiguities”. In: *41st IEEE/ACM International Conference on Automated Software Engineering Workshops*. ASEW ’26. Munich, Germany: Association for Computing Machinery, 2026. URL: https://doi.org/10.1145/3844136.3845868
Note : Workshop at ASE conference (A*)

2. Haitam El Hayani, Jolan Philippe, and Stephanie Challita. “Poliac: Production Observability for LLM-based Infrastructure as Code”. In: *41st IEEE/ACM International Conference on Automated Software Engineering Workshops*. ASEW ’26. Munich, Germany: Association for Computing Machinery, 2026. URL: https://doi.org/10.1145/3844136.3845869
Note : Workshop at ASE conference (A*)

3. Frédéric Loulergue, Jolan Philippe, and Nicolas Paul. “mech-uvl: A Mechanized Semantic Toolkit for the Universal Variability Language”. In: *Tenth International Workshop on Languages for Modelling Variability (MODEVAR@VARIABILITY 2026)*. VARIABILITY ’26. Limassol, Cyprus: Association for Computing Machinery, 2026. ISBN: 9798400720246
Note : Workshop at Variability conference (A)

4. Farid Arfi, Hélène Coullon, Frédéric Loulergue, Jolan Philippe, and Simon Robillard. “A Maude Formalization of the Distributed Reconfiguration Language Concerto-D”. in: ICE’24 (2024). URL: https://inria.hal.science/hal-04572043
Note : Workshop at DisCoTec conference

5. Frédéric Loulergue and Jolan Philippe. “Towards Verified Scalable Parallel Computing with Coq and Spark”. In: *Proceedings of the 25th ACM International Workshop*

on *Formal Techniques for Java-like Programs*. FTfJP 2023. Seattle, WA, USA: Association for Computing Machinery, 2023, pp. 11–17. URL: https://doi.org/10.1145/3605156.3606450
Note : Workshop at ECOOP conference (A*)

6. Jolan Philippe, Hélène Coullon, Massimo Tisi, and Gerson Sunyé. "Towards Transparent Combination of Model Management Execution Strategies for Low-Code Development Platforms". In: *Proceedings of the 23rd ACM/IEEE International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings*. MODELS '20. Virtual Event, Canada: Association for Computing Machinery, 2020. URL: https://doi.org/10.1145/3417990.3420206
Note : Workshop at MODELS conference (A)

3.5.4 Tutorial papers at international conferences with published proceedings

1. Baptiste Jonglez, Matthieu Simonin, Jolan Philippe, and Sidi Mohammed Kaddour. "Multi-provider Capabilities in EnOSlib: Driving Distributed System Experiments on the Edge-to-Cloud Continuum". In: *Distributed Applications and Interoperable Systems: 25th IFIP WG 6.1 International Conference, DAIS 2025, DisCoTec 2025, Lille, France, June 16–20, 2025*. Berlin, Heidelberg: Springer-Verlag, 2025, pp. 25–42. ISBN: 978-3-031-95727-7. URL: https://doi.org/10.1007/978-3-031-95728-4_2

3.5.5 Thesis

1. Jolan Philippe. "Contribution to the Analysis of the Design-Space of a Distributed Transformation Engine". PhD thesis. IMT Atlantique, 2022. URL: https://doi.org/10.5281/zenodo.8192984

2. Jolan Philippe. "Systematic development of Efficient programs on Parallel data structures". Northern Arizona University, 2019. URL: https://www.proquest.com/openview/02f765dad0386577b4d338b8e9abbad4

3.5.6 Project deliverables

1. Quentin Guilloteau, Ouadie Khebbeb, Éloi Perdereau, Jolan Philippe, Hélène Coullon, and Philippe Merle. *A Survey on Infrastructure as Code*. Produced for the TARANIS project: Model, Deploy, Orchestrate, and Optimize Cloud Applications and Infrastructure. 2026

2. Benedek Horváth, Jolan Philippe, Apurvanand Sahay, and Qurat ul ain Ali. *Multi-paradigm Distributed Transformation Engine*. Publicly produced for the project Training the Next Generation of Experts in Scalable Low-Code Engineering Platforms (Lowcomote), funded by Marie Skłodowska-Curie Actions (project number 813884). 2022. URL: https://www.lowcomote.eu/data/deliverables/D54.pdf

3. Sorour Jahanbin, Qurat ul ain Ali, Jolan Philippe, and Benedek Horváth. *Scalable Low-Code Artefact Persistence and Query.* Publicly produced for the project Training the Next Generation of Experts in Scalable Low-Code Engineering Platforms (Lowcomote), funded by Marie Skłodowska-Curie Actions (project number 813884). 2022. URL: https://www.lowcomote.eu/data/deliverables/D53.pdf
4. Benedek Horváth, Jolan Philippe, and Apurvanand Sahay. *Concepts for multi-paradigm distributed transformation.* Publicly produced for the project Training the Next Generation of Experts in Scalable Low-Code Engineering Platforms (Lowcomote), funded by Marie Skłodowska-Curie Actions (project number 813884). 2020. URL: https://www.lowcomote.eu/data/deliverables/D52.pdf

3.5.7 Extended abstracts and posters

1. Éloi Perdereau, Daniel Sokolowski, Jolan Philippe, Hélène Coullon, and Guido Salvaneschi. *Verifying the Stability of Terraform Executions.* Munich, Germany, 2026
2. Jolan Philippe and Frédéric Loulergue. "Towards Automatically Optimizing PySke Programs". In: *2019 International Conference on High Performance Computing and Simulation (HPCS).* 2019, pp. 1045–1046. URL: https://doi.org/10.1109/HPCS48598.2019.9188160
3. Jolan Philippe and Frédéric Loulergue. "Towards the Generation of Correct Java Programs (Research Poster)". In: *2018 International Conference on High Performance Computing and Simulation (HPCS).* 2018, pp. 1055–1056. URL: https://doi.org/10.1109/HPCS.2018.00166
4. Jolan Philippe, Wadoud Bousdira, and Frédéric Loulergue. "Formalization of a Big Graph API in Coq". In: *2017 International Conference on High Performance Computing and Simulation (HPCS).* 2017, pp. 893–894. URL: https://doi.org/10.1109/HPCS.2017.140

3.5.8 Diverse

1. Jolan Philippe. *Pédagogie d'un maître de conférences - Écrit réflexif.* Produced in the context of MCF trainings. 2026. URL: https://jolanphilippe.github.io/docs/publications/2026_relf_philippe.pdf
2. Jannik Laval, Jolan Philippe, Eric Cariou, Rabea Ameur-Boulifa, Sylvain Guérin, Olga Kouchnarenko, and Nawal Guermouche. *Défi Adaptation Continue (ADDYCT).* Produced in the context of GDR GPL renewal for 2026-2030. 2025. URL: https://jolanphilippe.github.io/docs/publications/2025_laval_gdr.pdf
3. Jolan Philippe. *Evaluation of Combinations of Model Management Execution Strategies for Low-Code Development Platforms.* 2021. URL: https://doi.org/10.5281/zenodo.10126465

4. Jolan Philippe. *Digital Signatures using Elliptic Curve with Extended Galois Fields*. 2019. URL: https://inria.hal.science/hal-03709246/

5. Jolan Philippe and Gowanlock Michael. *2D-Clustering through Approximation Method using Geometric Calculation*. 2018. URL: https://doi.org/10.5281/zenodo.10126470

4 Scientific activities

4.1 Participation in projects

- I lead the **UniVerSE** (Unification and Verification of configuration SpacEs) project, which aims to unify existing work on configuration spaces. The Universal Variability Language (UVL) was initially proposed as a standard language for defining the configuration spaces of software product lines, but it is now increasingly used to model variability in a broader range of systems. Despite this growing adoption, UVL remains fragmented: its semantics can vary depending on the context in which it is used and on the specific implementation. Moreover, UVL offers limited expressiveness for specifying constraints and does not fully leverage the extensive body of work developed by the constraint-programming and constraint-reasoning communities. With UniVerSE, I aim to strengthen UVL by extending its foundations and providing a unified formal semantics for the specification and verification of configuration spaces.

- I am a co-leader of the **RaDyD** (Resilience through Dynamic Adaptation for Data Processing) project. The purpose of the project is to investigate how data-processing systems can remain operational when their execution environment is degraded. The project explores semantics-aware resilience, where systems dynamically reconfigure their software components, execution strategies, or hardware resources while preserving the strongest possible semantic guarantees. In particular, RADyD studies how semantic contracts can characterize substitutions as exact, conditionally equivalent, or approximate with explicit quality bounds, turning adaptation into controlled and explainable degradation rather than ad hoc fallback.

- I am a part of the ANR **For-Coala** project. The project aims to bring formal guarantees to Infrastructure-as-Code (IaC) languages used to configure and reconfigure distributed software systems. The project seeks to bridge the gap between widely adopted DevOps technologies and formally defined academic approaches, focusing in particular on Ansible and the academic language Concerto. Its goal is to develop mechanized formal semantics for these languages and verified, semantics-preserving transformations between them, ultimately providing open-source, formally certified configuration languages, execution engines, and translation tools.

- As a postdoctoral researcher in the **Taranis** project under the **PEPR Cloud** initiative, I focused on redesigning the techniques for developing and verifying Infrastructure as Code (IaC) languages. The project aims to simplify cloud application development. My work addresses tooling gaps for declarative languages, particularly in type systems, linters, and IDEs. These tools are crucial for making informed decisions about topology and resource allocation, considering DevOps (operational efficiency), SecDevOps (security), FinOps (financial optimization), and GreenOps (ecological impact).

- As a postdoctoral fellow for the **ANR SeMaFoR** project, I contributed to WP3: Decentralized Reconfigurations. The SeMaFoR project aimed to bring significant advances in the collaborative exploitation of Fog resources. In my postdoctoral position, I was responsible for the task related to the inference and planning of reconfiguration programs.

- My thesis was funded by the **Marie Skłodowska-Curie Actions**, through the **Lowcomote** project. Lowcomote was a European project that aimed to train a generation of professionals in the design, development, and operation of low-code platforms. In this project, I participated in the writing of several deliverables, as well as in the general coordination of the project by volunteering to represent the project's doctoral students and by contributing to the administration of its website.

- I was a research engineer in the **Girafon** project, funded by the Centre-Val de Loire Region (**APR-IA**), which aimed to propose algorithms for querying, exploiting, and analyzing large graphs using rich query languages and distributed computing. I contributed to the task "Structured Parallel Programming on Large Graphs".

4.2 Supervision

- **[2026-11-01, 2029-10-31] Nathan Fouéré, Ph.D candidate, Taranis project.** Empirical and formal study of the resilience of decentralized Infrastructure-as-Code solutions.

- **[2025-09-01, 2028-08-31] Olivia Proust, Ph.D candidate, For-Coala project.** Formalization and verification of Infrastructure as Code (IaC) scripts for configuration management (Ansible).

- **[2026-05-18, 2026-08-28] Paul Nicolas, M1 internship, UniVerSE project.** Exploration of UVL (Universal Variability Language) ecosystem with its different semantics.

- **[2024-09-01, 2027-08-31] Haitam El Hayani, Ph.D candidate, Taranis project**. Extending Infrastructure as Code (IaC) languages, focusing on enhancing their capabilities and improving their integration with modern cloud deployment practices.

4.3 Complete list of talks

Presentations at conferences and workshops with international audiences

- "Towards Semantics-Aware Resilience in Composable Data Systems", September 30^{th}, 2026. European Conference on Advances in Databases and Information Systems (ADBIS), Orléans (France)
- "Fast Choreography of Cross-DevOps Reconfiguration with Ballet: A Multi-Site OpenStack Case Study", March 13^{th}, 2024. International conference: Software Analysis, Evolution and Reengineering (SANER), Rovaniemi (Finland)
- "Multi-Paradigm Distribution for Model Management Operations", December 15^{th}, 2022. Lowcomote Industrial Workshop, Online
- "Executing Certified Model Transformations on Apache Spark", October 17^{th}, 2021. International conference : Software Language Engineering (SLE), Online
- "Towards Transparent Combination of Model Management Execution Strategies for Low-Code Development Platforms", October 19^{th}, 2020. Workshop international : Low-code, International Conference on Model Driven Engineering Languages and Systems (MODELS), Online
- "Automatic Optimization of Python Skeletal Parallel Programs", December 9^{th}, 2019. Internationale conference : Algorithms and Architectures for Parallel Processing (ICA3PP), Deakin University - Melbourne (Australia)
- "New List Skeletons for the Python Skeleton Library", December 6^{th}, 2019. International conference: Parallel and Distributed Computing, Applications and Technologies (PDCAT), Gold Coast (Australia)
- "Towards Automatically Optimizing PySke Programs", July 18^{th}, 2019. Poster session, Internationale conference : High Performance Computing & Simulation (HPCS), Dublin (Ireland) [poster]
- "PySke: Algorithmic Skeletons for Python", July 16^{th},. Internationale conference : High Performance Computing & Simulation (HPCS), Dublin (Ireland)
- "Coq2Java : From MiniML to Java code, Formalization of a compiler", July 17^{th}, 2018. International conference: High Performance Computing & Simulation (HPCS), Orléans (France) [poster]
- "Verified Skeletons on Trees", July 13^{th}, 2018. Poster session, International conference: High-Level Parallel Programming and Applications (HLPP), Orléans (France) [poster]

- "Formalization of a Big Graph API in Coq", July 20^{th}, 2017. Poster session, International conference: High Performance Computing & Simulation (HPCS), Genoa (Italia) [poster]

Presentations in seminars and working groups

- "Défi ADDyCT: ADaptation Dynamique et ConTinue", June 4^{th}, 2026. Congrès national des sciences du logiciel, Lille (France)
- "Kick-off du défi ADaptation DYnamique et ConTinue - ADDYCT", April 29^{th}, 2026. Online
- "SeMaFoR WP3 - Results", October 7^{th}, 2025. IMT Atlantique, Nantes (France)
- "Systèmes Logiciels Adaptables", July 7^{th}, 2025. Journées d'équipe LMV, La Ferté-Saint-Aubin (France)
- "Managing Conflicts in Cross-DevOps Declarative Reconfigurations", March 24^{th}, 2025. Taranis Webinar, Online
- "SyLA - Systèmes Logiciels Adaptables, ", March 3^{st}, 2025. Défi GDR-GPL 2026 - 2030, Université Sorbonne (LIP6), Paris (France)
- "Correction in distributed systems, from usage to reconfiguration - Progress and insights from my research journey", January 24^{th}, 2025. STR team seminar, Ecole Centrale, Nantes (France)
- "Towards Management of Unsatisfiable Reconfiguration in Ballet", October 7^{th}, 2024. Taranis project plenary, Centre Inria de Paris (France)
- "Fast Choreography of Cross-DevOps Reconfiguration with Ballet, Multi-Site OpenStack Case Study", June 5^{th}, 2024. Journées GDR GPL, Université Strasbourg (France)
- "Fast Choreography of Cross-DevOps Reconfiguration with Ballet and resolving SAT problems with a decentralized approach", April 8^{th}, 2024. SeMaFoR project plenary, Université Sorbonne - Paris (France)
- "Fast Choreography of Cross-DevOps Reconfiguration with Ballet, Multi-Site OpenStack Case Study", December 13^{th}, 2023. Working group in VELVET days, Nantes (France)
- "Fast Choreography of Cross-DevOps Reconfiguration with Ballet, Multi-Site OpenStack Case Study", November 9^{th}, 2023. STACK team seminar, Guérande (France)
- "SeMaFoR Project and Planning for Decentralized Reconfiguration", October 26^{th}, 2023. TASC team seminar, IMT Atlantique - Nantes (France)

- “SeMaFoR Project and Concerto-D for decentralized reconfiguration of Fog systems”, July 3^{rd}, 2023. Naomod team seminar, IMT Atlantique - Nantes (France)
- “SeMaFoR Project and Decentralized reconfiguration plan synthesis”, June 6^{th}, 2023. DAPI department seminar, Nantes (France)
- “Decentralized reconfiguration plan synthesis”, April 20^{th}, 2023. SeMaFoR project plenary, IMT Atlantique - Nantes (France)
- “Contribution to the Analysis of the Design-Space of a Distributed Transformation Engine”, January 23^{th}, 2023. LMV team seminar (Langages, Models et Verification) - Lifo, Université d'Orléans (France)
- “Programming Models and Distributed systems”, November 18^{th}, 2019. STACK team seminar, Université de Nantes (France)
- “PySke: Algorithmic Skeletons for Python”, October 18^{th}, 2019. Working group in GDR LAHMA, LACL - Créteil (France)
- “Algorithmic Skeletons for Distributed Computing”, November 16^{th}, 2018. Graduate student seminar, Northern Arizona University - Flagstaff, AZ (USA)
- “BMF and Parallelism”, May 25^{th}, 2018. Speech at Summer School on Formal Methods (SSFT), Menlo College - Atherton (USA)
- “Formalization of a Big Graph API in Coq”, September 13^{th}, 2017. Northern Arizona STEM Poster Session, Northern Arizona University - Flagstaff, AZ (USA) [poster]

Defenses

- “Contribution to the Analysis of the Design-Space of a Distributed Transformation Engine”, December 19^{th}, 2022. Doctoral thesis defense, IMT Atlantique - Nantes (France)
- “Systematic development of Efficient programs on Parallel data structures”, May 3^{th}, 2019. Master thesis defense, Northern Arizona University - Flagstaff, AZ (USA)

4.4 Relectures

During my thesis and my postdoctoral work, I had the opportunity to participate, as a reviewer, in the evaluation of scientific work submitted to international conferences and journals. The table 4 gives an overview of my participation.

Year	Name	Description
2026	UCC 2026	International Conference on Utility and Cloud Computing
2025	BDCAT 2025	International Conference on Big Data Computing, Applications and Technologies
	UCC 2025	International Conference on Utility and Cloud Computing
	SPLC 2025	International Systems and Software Product Line Conference
	DebConf 2025	International Debian Conference
	Lowcode 2025	International Workshop on Modeling in Low-Code Development Platforms
	ICCS 2025	International Conference on Conceptual Structures
2024	SoSyM	International Journal on Software and Systems Modeling
2023	COLA	Journal of Computer Languages
	SBAC-PAD 2023	International Symposium on Computer Architecture and High Performance Computing
	CP 2023	International Conference on Principles and Practice of Constraint Programming
2021	MODELS 2021	International Conference on Model Driven Engineering Languages and Systems
2020	MODELS 2020	International Conference on Model Driven Engineering Languages and Systems
	ECMFA 2020	European Conference on Modelling Foundations and Applications
	ICCS 2020	International Conference on Computational Science

Table 4: List of reviews for international scientific journals or conferences

4.5 Collective responsabilities

2026 - 2030	**Chair of the GDR SciLog challenge ADDYCT** (ADaptation DYnamique et ConTinue), with Jannik Laval (Université Lumière Lyon 2), and Nawal Guermouche (LAAS-CNRS, Toulouse)
Nov 2025 - Oct 2026	**Facilities Coordinator** of the building IIIA for LIFO collaborators
Feb 2025 - June 2026	**Chair of SyLA (Système Logiciel Adaptable) working group** at GDR SciLog. With Rabea Ameur-Boulifa (Telecom Paris), and Simon Blyudze (Université de Lille)
Dec 2023	**Help with organizing VELVET days** (Verification Software Engineering for DevOps and Reconfiguration) at IMT Atlantique - Nantes (France)
Since Feb 2023	**Site manager and administrator for STACK** team [site]
July 2022	**Student volunteer** for the STAF conference in Nantes (France) https://staf2022.univ-nantes.io/
Sep 2019 - Mar 2023	**Site manager and administrator for Lowcomote** project [site]
Oct 2019 - Jan 2021	**Representative of the Lowcomote Ph.D. students**
Sep 2019 - Mar 2023	**Site manager and administrator for Naomod** team [site]
Nov 2018 - Août 2019	**Funder and main organizer of "SICCS graduate student seminar"** at Northen Arizona University - Flagstaff AZ (USA)
Juil 2018	**Student volunteer** for the HPCS conference in Orléans (France) https://hpcs2018.cisedu.info/

5 Additional parts

This file concludes with the reports from the thesis referees and the qualification certificate, to be "Maitre de conférences" (Section 27)

Conclusion of the study of the manuscript for a defense

Doctorant :

M. Jolan PHILIPPE Département IMT Atlant que : DAPI

Int tulé de la thèse : Mult -Paradigm Distribut on for Model Management Operat ons

Spécialité : Informat que

Directeur de thèse : M. Gerson SUNYE

Date de soutenance prévue : 19/12/2022

Rapporteur :

(Civilité Prénom - NOM) : M. Jesus SANCHEZ CUADRADO Titre : Associate professor

Etablissement : Université de Murcia - Espagne

☒ If this thesis does not need any modifications, the thesis can be defended.

☐ A few changes need to be made to this dissertation; the defense of this work can proceed without resubmitting this document for examination.

☐ Significant changes must be made to this thesis.

☐ The thesis cannot be defended as is; the new version will have to be resubmitted for examination.

Dated at ....Murcia........ le ....7 of Dec. 2022........

Signature

Evaluation Report on Mr Jolan Philippe's PhD Thesis, entitled "Contribution to the Analysis of the Design-Space of a Distributed Transformation Engine"

This report evaluates the PhD Thesis of Mr Jolan Philippe, entitled "Contribution to the Analysis of the Design-Space of a Distributed Transformation Engine". The context of this thesis is Model-Driven Engineering (MDE), in particular the development of model transformation engines and their associated characteristics. The thesis is focused on studying how the different choices available in the design of a distributed model transformation engine affect to its characteristics, in particular trying to address the trade-off between correctness and performance. The thesis has produced so far two articles, in an international conference and in a workshop. The conference paper was presented at SLE'21, which is a well-known conference in the modeling community. Moreover, the thesis has produced a set of software artifacts and experiments, which are available as open source.

The topic of the thesis is relevant for the modeling community since there is a need to build better transformation engines (e.g., more efficient) but also keeping in mind the need to guarantee correctness and to allow developers reason about their code (e.g., prove properties of the transformation).

Chapter 2 presents some preliminary information which is needed to set the context of the work and Chapter 3 presents a detailed and comprehensive review of the state-of-the-art related to the topics of the thesis. In particular, three main topics have been studied in depth. First, works related to the efficiency of model transformations are classified according to whether they are focused on model queries or transformations. Another classification dimension is whether the matching phase of a transformation execution is optimized or not. The final classification dimension is related to works addressing distribution. After this, the study of the state-of-the-art moves to works addressing the correctness of model transformations, which will be lated combined with efficiency approaches in the following works (i.e., by building a provably correct transformation engine). Finally, a few works related to the applying variable strategies to improve the performance of model transformations are discussed.

The core work of the thesis is reported in chapters 4, 5 and 6, which are discussed in the following.

Chapter 4 discusses several strategies which can be used to implement distributed model queries (which are an essential part of a model transformation engine which wants to support complex transformations) and establishes a mapping between OCL and Scala/Spark. The study is done by reusing a well-known benchmark of the model transformation community (the TTC'18 case) and implementing the same query in Spark (with a direct translation), Pregel and Map-Reduce. The algorithmic complexity of each implementation is studied analytically, so that the weaknesses and strengths of each one is derived with the idea of motivating the need of transformation engines which are able to determine which strategy is better according to the execution context and the input models. Finally, the implementations are evaluated empirically by running the transformations using different sets of input models on a shared memory machine with an Intel Core i7-8650U having 8 cores at 1.90GHz and a memory of 32GB. The results seem to confirm that no single strategy suits all use cases, which reinforces the idea of having engines with the ability of selecting the best strategy at runtime.

Chapter 5 is the main contribution of the thesis. It presents an evolution of the CoqTL transformation language to favor parallelism. The evolution implements a new version of the main execution function of CoqTL in which the phases are organized in a way that favors parallelism. This part of the thesis is illustrated with a concrete example, a transformation from relational models to class diagrams. Then, this new version of CoqTL is proved to be semantically equivalent to the original. It is interesting that the proof effort is measured, which can also be used by others as a kind of benchmark. On the other hand, to evaluate the performance of this approach, programs written in CoqTL are translated into Scala (and Spark). This is done manually, and the development of a compiler is left for future work (although this would have been a very good addition to the thesis). Using the proposed translation, the performance of the engine is evaluated on a cluster by analyzing the effect of the design decisions in each of the different phases of the transformation engine execution.

Chapter 6 concludes the technical contribution of the thesis with a study of the different implementation options for the transformation engine. The chapter is illustrated with a well known case study proposed in TTC'14, which has been used by several other works to analyze distribution and parallelism of model transormation. In this part of the thesis, instead of a one-to-one mapping from CoqTL to Scala, different execution strategies are identified and tested in the transformation engine to study its effect in the performance of the engine. The different configurations have been formalized with a feature model and have been implemented as modules of the transformation engine.

Altogether, this thesis performs the first steps in the study of the different aspects that encompass the design and implementation of efficient and correct distributed model transformation engines. In this sense, although a number of interesting enhancements of this work are left as future work, this is a good step towards addressing them and thus improve the state-of-the-art in this topic.

Therefore, I acknowledge that this work complies with the requirements for a doctoral degree and I am in favor of Jolan Philippe's PhD defense.

Jesús Sánchez Cuadrado
Dpto. de Informática y Sistemas, Universidad de Murcia, España

Conclusion of the study of the manuscript for a defense

Doctorant :

M. Jolan PHILIPPE — Département IMT Atlantique : DAPI

Intitulé de la thèse : Multi-Paradigm Distribution for Model Management Operations

Spécialité : Informatique

Directeur de thèse : M. Gerson SUNYE

Date de soutenance prévue : 31/12/2022

Rapporteur :

(Civilité Prénom - NOM) : M. Mathias TICHY — Titre : Professeur

Etablissement : ULM University Allemagne

☒ If this thesis does not need any modifications, the thesis can be defended.

☐ A few changes need to be made to this dissertation; the defense of this work can proceed without resubmitting this document for examination.

☐ Significant changes must be made to this thesis, The thesis cannot be defended as is; the new version will have to be resubmitted for examination.

Made at Ulm le 28/11/2022

Signature

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28.11.2022

Review of the PhD Thesis of Jolan Philippe
"Contribution to the Analysis of the Design Space of a Distributed Transformation Engine"

Context and Aim of the Thesis

Model Transformation are key artefacts in the area of model-driven engineering. They are used to transform a single model or create a target model from a source model. There exist many different model transformation languages like ATL, Henshin, Viatra. Historically, programs written in those model transformation language have been executed locally in a single thread by the corresponding engine. However, the size of models has been increasing steadily with papers reporting about models bigger than 100k objects and the TTC movie benchmark (published in 2014) containing over 2.5m objects. This gave rise to several approaches that execute model transformations in a parallel and/or distributed fashion aiming for vertical and/or horizontal scalability.

The aim of the thesis by Jolan Philippe is threefold:

1. A manual transformation of model queries – specifically OCL queries – to different distributed programming paradigms/platforms, e.g., Spark and Pregel.
2. A parallel version of the CoqTL transformation language that has been formally proven to be input/output equivalent compared to the standard CoqTL version using the Coq theorem prover.
3. A manual transformation of the parallel CoqTL programs to Spark implementations and the evaluation of different implementation as well as configuration variants.

Structure and Contents

The thesis contains seven chapters with 108 pages, two appendices and an extensive bibliography. Contents of the thesis have been published in two papers – one at the Software Language Engineering conference (2021) and one at the LowCode workshop (2020). Jolan Philippe is first author of both papers.

The first chapter introduces the area of model-driven engineering including model transformations and sketches the three contributions of the thesis.

Jolan Philippe introduces the necessary foundations of his research in the second chapter. This includes a nice overview about models and model transformations – particularly the Atlas Transformation Language, concepts of distributed computing – particularly the Spark platform –, and interactive theorem proving – particularly the Coq proof assistant. The chapter covers the necessary contents in a systematic and coherent manner.

The following Chapter 3 contains a discussion of the state of the art. The first and major part of the chapter reviews the different existing approaches to improve efficiency of model transformation by exploiting task parallelism and data parallelism as well as distributed asynchronous computation. This section properly discusses the different major approaches like the ones by Viatra, IncQuery, Henshin, ATL that are using different paradigms like RETE-networks, the Bulk Synchronous Parallel abstract computer. The latter two sections of the chapter shortly discuss semantics and proving of model transformations as well as feature models to specify the configuration space used for benchmarking.

The following three chapters cover each one of the three contributions as mentioned above. Jolan Philippe presents a distributed implementation of OCL queries base on the Spark platform in Chapter 4. He systematically presents different implementation variants of OCL queries using the different concepts and evaluates them on 6 different data sets. The results show that the size of the dataset has an impact on the scalability of the different implementation variants. It would have been interesting to show in more depth the full set of mapping rules from OCL to the different variants as well as to investigate systematically which properties of the data set affecting the scalability of the different implementations noting that bigger data sets, e.g., data set 5, appear to scale less than smaller data sets, e.g., data set 4.

Chapter 5 presents a parallel version of CoqTL – a transformation language which supports the interactive verification of transformations against contracts. The parallelizable CoqTL transformation is then translated to Scala and executed on the Spark platform. Jolan Philippe presents the refinement proof for parallelizable CoqTL and sketches the translation to Scala and Spark using examples. The evaluation on rather small models (up to 700 elements) shows that the solution scales rather bad. Jolan Philippe correctly identifies that small computations are (at least one of) the reason for the bad scalability and shows that by introducing different sleep times to emulate more complex calculations the scalability can improve. Valuable next steps based on the results would be a systematic and in-depth description of the translation to Scala and Spark, an evaluation with bigger models, and a translation to Scala that tries to improve the scalability problem, e.g., by grouping computations into bigger chunks to reduce the communication overhead.

Finally, the third major chapter (Chapter 6) covers the presentation and evaluation of different implementation alternatives. This includes for example how to navigate over references between nodes on the input model or how trace links between input and output model elements are handled. The different variants are shown in three feature diagrams with 28 (leaf) features. Jolan Philippe focuses mostly on link navigation and trace management alternatives in the discussion and illustrates those with two code snippets. The evaluation (on models up to 10.000 elements) shows that using HashMaps hugely outperforms the List implementations on the two features "DynamicLinksNavigation" and "TraceLinksNavigation". Reasonable next steps would be to systematically present and evaluate the variants as well as evaluate them on bigger models.

The thesis concludes with a short synthesis as well as a good and extensive overview on limitations and future research directions summarizing the results of the three major chapters in chapter 7.

Summative Assessment

In summary, Jolan Philippe presents a concise PhD thesis which presents an original contribution to the state of research of parallel model transformations. He shows a solid knowledge of the state of the art in the model transformations research field and the ability to independently conduct research. Hence, I believe that the thesis meets the standards required for the degree of a PhD and, thus, I recommend that Jolan Philippe proceeds to defend his PhD thesis.

Matthias Tichy

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RECAPITULATIF DES QUALIFICATIONS VALIDES
publiées au 08/03/2024*

Par M. JOLAN PHILIPPE

Né(e) le 27/09/1995 à GIEN

CORPS	SECTION**	N° QUALIFICATION	DATE	CAMPAGNE	DATE EFFECTIVE PEREMPTION
MCF	27	24227403526	09/02/2024	2024	31/12/2028

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(**)L'intitulé correspondant au numéro des sections est consultable sur le portail GALAXIE.

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